

# PAPER\_895: E(t) Vs Quintessence Scalar Field Contrast

**Author:** Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** EtVsQuintessenceScalarFieldContrastCalc (CP4 #479) **CVW:** v2.0.0 compliant

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## Abstract

Comparison of UQFF E(t) with quintessence scalar field dark energy. Quintessence uses  $V(\varphi) = M^4 \cdot \exp(-\lambda\varphi/M_{Pl})$  with slow-roll dynamics. UQFF uses  $S_{26}$  Ramanujan  $\times \exp(\kappa t)$  with 2 calibrated parameters. 10-row contrast table covering field content, potential, EOS, free parameters, time evolution, vacuum, lab testability, GW impact, fine-tuning, and origin.

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## 1. Core Equations

$V(\varphi) = M^4 \cdot \exp(-\lambda\varphi/M_{Pl})$   
 $w_{quint} = (KE - V)/(KE + V)$   
10-aspect comparison

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## 2. Parameters

| Parameter    | Default | Description                    |
|--------------|---------|--------------------------------|
| phi_0        | 1e-5    | Initial scalar field value     |
| phi_dot_0    | 1e-40   | Initial field velocity         |
| lambda_quint | 1.0     | Quintessence coupling constant |
| M_scale      | 1e-3 eV | Quintessence mass scale        |

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## 3. UQFF Integration

This calculator integrates et\_scm\_vacuum.py from Session 207 into the CP4 pipeline as class #479.

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